

# Information Theory: Worked Solutions

## Problem 1. Entropy, cross-entropy, and their gap

**Problem.** For  $p = (1/2, 1/3, 1/6)$ , compute entropy in bits, cross-entropy against the uniform law on the same three symbols, and KL divergence to that law. Interpret their difference.

**Solution.** Let  $q = (1/3, 1/3, 1/3)$ . The subscript 2 denotes base-two logarithms and hence bit units. Direct substitution gives

$$\begin{aligned} H_2(p) &= -\sum_{i=1}^3 p_i \log_2 p_i = \frac{1}{2} + \frac{1}{3} \log_2 3 + \frac{1}{6} \log_2 6 \\ &= \frac{2}{3} + \frac{1}{2} \log_2 3 \approx 1.45915 \text{ bits}, \\ H_2(p, q) &= -\sum_{i=1}^3 p_i \log_2(1/3) = \log_2 3 \approx 1.58496 \text{ bits}. \end{aligned}$$

The expected log-ratio is

$$\begin{aligned} D_{\text{KL},2}(p||q) &= \sum_i p_i \log_2 \frac{p_i}{q_i} = \frac{1}{2} \log_2 \frac{3}{2} + \frac{1}{3} \log_2 1 + \frac{1}{6} \log_2 \frac{1}{2} \\ &= \frac{1}{2} \log_2 3 - \frac{2}{3} \approx 0.12581 \text{ bits}. \end{aligned}$$

Thus  $H_2(p, q) - H_2(p) = D_{\text{KL},2}(p||q)$ . Entropy is the expected logarithmic cost under the true law  $p$ ; the gap is the extra expected cost of assigning uniform probabilities  $q$ . These are ideal log costs; individual binary prefix-code lengths also have integer constraints.

## Problem 2. A perfect copy and an independent variable

**Problem.** Let  $X$  be uniform on  $K$  symbols, with  $K \geq 1$ . First set  $Y = X$ , then compare with a  $Y$  that is independent of  $X$  and uniform on the same alphabet. In each case find  $H(X, Y)$ ,  $H(Y | X)$ , and  $I(X; Y)$ .

**Solution.** Use natural logarithms, so the answers are in nats. If  $Y = X$ , the joint law has  $K$  positive cells  $(x, x)$ , each of probability  $1/K$ . Hence

$$H(X, Y) = \log K.$$

Given  $X = x$ ,  $Y$  is known exactly, so  $H(Y | X) = 0$ . Since  $Y$  is itself uniform,

$$I(X; Y) = H(Y) - H(Y | X) = \log K.$$

For an independent uniform  $Y$ , all  $K^2$  joint cells have probability  $1/K^2$ . The conditional law of  $Y$  is its unchanged uniform marginal, giving

$$H(X, Y) = 2 \log K, \quad H(Y | X) = \log K, \quad I(X; Y) = 0.$$

In the first case knowledge of  $X$  removes all uncertainty about  $Y$ ; in the second it removes none. Replacing log by  $\log_2$  expresses each answer in bits. For  $K = 1$  all quantities are zero.

## Problem 3. Information in a noisy copy

**Problem.** Let  $X \sim \text{Bernoulli}(1/2)$  and let  $N \sim \text{Bernoulli}(\varepsilon)$  be independent of  $X$ . Define  $Y = X \text{ xor } N$ . At  $\varepsilon = 1/4$ , calculate the four joint probabilities and recover  $I(X; Y)$  directly from the KL definition.

**Solution.** A matching pair requires  $N = 0$ , and a mismatching pair requires  $N = 1$ . Thus

|         | $Y = 0$ | $Y = 1$ | $p_X$ |
|---------|---------|---------|-------|
| $X = 0$ | 3/8     | 1/8     | 1/2   |
| $X = 1$ | 1/8     | 3/8     | 1/2   |
| $p_Y$   | 1/2     | 1/2     | 1     |

Both marginals are fair, so their product assigns probability 1/4 to every cell. Using natural logarithms,

$$\begin{aligned}
 I(X; Y) &= \sum_{x,y} p(x,y) \log \frac{p(x,y)}{p_X(x)p_Y(y)} \\
 &= 2 \frac{3}{8} \log \frac{3/8}{1/4} + 2 \frac{1}{8} \log \frac{1/8}{1/4} \\
 &= \frac{3}{4} \log \frac{3}{2} + \frac{1}{4} \log \frac{1}{2} = \frac{3}{4} \log 3 - \log 2 \approx 0.13081 \text{ nats.}
 \end{aligned}$$

Dividing by  $\log 2$  gives approximately 0.18872 bits. This agrees with

$$I(X; Y) = \log 2 - h(1/4), \quad h(t) = -t \log t - (1-t) \log(1-t),$$

because the flip leaves  $h(1/4)$  nats of conditional uncertainty about  $Y$  after  $X$  is known.

## Problem 4. Shifting log weights

**Problem.** For a finite vector  $a \in \mathbb{R}^K$ ,  $K \geq 1$ , define  $\text{LSE}(a) = \log \sum_{k=1}^K e^{a_k}$ . Show that  $\text{LSE}(a + c\mathbf{1}) = c + \text{LSE}(a)$ . Derive invariance of normalized log probabilities and check it for  $c = 10^4$ .

**Solution.** Factoring out the common multiplier  $e^c$  gives the exact identity

$$\text{LSE}(a + c\mathbf{1}) = \log \left( e^c \sum_k e^{a_k} \right) = c + \text{LSE}(a).$$

Consequently, if  $\log q_k = a_k - \text{LSE}(a)$ , then

$$(a_k + c) - \text{LSE}(a + c\mathbf{1}) = a_k - \text{LSE}(a) = \log q_k.$$

To evaluate the identity without overflow, let  $m = \max_k a_k$  and use

$$\text{LSE}(a) = m + \log \sum_k e^{a_k - m}.$$

All exponentials are now at most one, and at least one equals one. For example, take  $a = (-2, 0, 1)$ :

```
import numpy as np

def lse(a):
    m = np.max(a)
    return m + np.log(np.exp(a - m).sum())

a = np.array([-2., 0., 1.])
c = 1e4
logq = a - lse(a)
shifted_logq = (a + c) - lse(a + c)
np.testing.assert_allclose(lse(a + c), c + lse(a))
np.testing.assert_allclose(shifted_logq, logq, rtol=0, atol=1e-11)
# logq is approximately [-3.349012, -1.349012, -0.349012].
```

The mathematical invariance is exact. Floating-point subtraction may introduce a small rounding difference; the check allows for it. Computing  $e^{a_k+c}$  directly would overflow for this shift even though the normalized probabilities are well defined.

## Problem 5. Differential entropy and a change of scale

**Problem.** Let  $X \sim \text{Unif}(0, 1/4)$  and  $Y = 8X$ . Find both differential entropies. Explain why a negative differential entropy does not contradict the discrete entropy bound.

**Solution.** The density of  $X$  is 4 on  $(0, 1/4)$ , so, in nats,

$$h(X) = - \int_0^{1/4} 4 \log 4 \, dx = -\log 4.$$

The variable  $Y$  is uniform on  $(0, 2)$ , with density  $1/2$ . Therefore

$$h(Y) = - \int_0^2 \frac{1}{2} \log \frac{1}{2} \, dy = \log 2.$$

The change is  $h(Y) - h(X) = \log 8$ , agreeing with  $h(aX) = h(X) + \log |a|$  for nonzero scalar  $a$ .

Discrete entropy is a weighted sum of  $-\log p(x)$ , which is nonnegative because each probability is at most one. A density is probability per unit length and can exceed one; consequently  $-\log f(x)$  can be negative. Differential entropy also changes when the coordinate unit changes, whereas a one-to-one relabeling of discrete outcomes preserves their entropy. For example, finely quantizing  $X$  into bins of width  $\Delta$  produces a discrete entropy approximately  $h(X) - \log \Delta$ , rather than  $h(X)$  itself. Thus the two quantities obey different nonnegativity and invariance rules.